

MADHAN MATTHEW S

Bengaluru, India | +91 78996 43760 | madhan4530@gmail.com | LinkedIn | GitHub | LeetCode

SUMMARY

AI & ML Engineering undergraduate with an 8.53 CGPA and hands-on experience across software development, data analytics, artificial intelligence, machine learning, NLP, and backend engineering. Proficient in Python, SQL, Java, JavaScript, modern web technologies, databases, and AI/ML frameworks, with experience building practical, data-driven and intelligent applications. Strong foundation in data structures, algorithms, databases, computer networks, and software engineering, seeking opportunities to apply technical and analytical skills to real-world problems and contribute to impactful technology solutions.

EDUCATION

CMR Institute of Technology, Bengaluru	Expected 2027
Bachelor of Engineering – Artificial Intelligence & Machine Learning CGPA: 8.53	
Christ Global Composite PU College, Bengaluru	2023
Pre-University Course (PCMB) 93.5%	
New Baldwin International School, Bengaluru	2021
10th Grade 86.3%	

TECHNICAL SKILLS

- Programming:** Python, Java, C, SQL, JavaScript
- AI / ML:** Machine Learning, NLP, Generative AI & LLMs, Transformers, spaCy, Sentence-Transformers, Pandas
- Web & Backend:** FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, React, D3.js, Tailwind CSS, HTML, CSS
- Databases:** MongoDB, PostgreSQL, MySQL
- Data & Visualization:** Tableau, Power BI, Excel
- Tools & DevOps:** Git, GitHub, Docker, Linux, Postman, Render, Netlify, Figma
- Core CS:** Data Structures & Algorithms, Computer Networks, Software Testing

PROJECTS

AI-Powered Multi-Agent Emergency Response & Traffic Coordination System (Ongoing Project)	Python, FastAPI, MongoDB
- Built a multi-agent system (Ambulance, Traffic, Police, Hospital, Coordinator) for incident triage, routing and resource allocation across Bengaluru. - Engineered ML-based severity classification and ETA prediction using OpenRouteService routing, Haversine fallback and route caching. - Implemented hospital selection by trauma capability, beds, specialties and distance, with FastAPI REST APIs and WebSocket dispatch updates; benchmarked 8 Bengaluru scenarios against a distance-based baseline.	
AI-Powered Inconsistency Detector for Police Case Analysis	Python, Flask, Groq API (LLaMA 4 Scout), spaCy
- Built a GenAI forensic tool that cross-examines witness statements, FIRs and CCTV summaries to flag contradictions in timelines, suspect appearance and vehicle details. - Designed a hybrid NLP pipeline combining LLM reasoning, semantic similarity (MiniLM-L6-v2), entity extraction and regex pattern matching. - Delivered a Flask REST backend with Simple and Detailed views (similarity scores, conflict breakdowns) and one-click ReportLab PDF export.	
GoalPulse – AI-Powered Goal Tracking Web App	React 19, D3.js, Framer Motion, Vite
- Built a full-stack React app on the MEMbrain semantic memory API to store, retrieve and semantically search goals and progress events in real time. - Created a D3.js force-directed "Goal Constellation" graph with physics simulation, zoom/pan and drag; managed state with React Context and custom hooks across 6+ components.	

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle University (2025)
- The Complete Full Stack Web Development – Udemy (2025)
- UI/UX Design Fundamentals – Udemy (2025)

ACTIVITIES

- Runner-up, hackathon conducted by Tovard's Club, CMRIT
- Participant: GenAI for Police (TrendyTech), MEMbrain (CMRIT), Odoo x NMIT 2025, InnovateX 2025, AgentX 2025
- Hobbies & Interests: Content Creation & Video Editing • Music & Singing • Dance
- Languages: English, Kannada, Hindi, Tamil, Telugu